# FLOOD RESILIENCE & RESPONSE PLATFORM

Technical Project Submission

**Project Purpose**
A web-based disaster-management platform connecting flood preparedness, community reporting, drone-based situational awareness, resource coordination, hazard-aware routing and shelter tracking.

## Core Focus: Response Coordination Engine

The platform is designed around a human-in-the-loop response workflow. Information from citizens and simulated drone observations is converted into an operational state, which is then used to recommend resources, volunteers and routes for authority approval.

**Submission Scope:** Technical proposal / MVP specification. Where implementation details are not yet completed, they are described as proposed rather than claimed as deployed.

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# 1. Executive Summary & Core Value Proposition

## 1.1 Executive Summary

Flood disasters create a coordination problem as much as a physical hazard problem. During a flood, information arrives from multiple sources, roads can become inaccessible, available resources change, citizens may be separated from family members, and responders must act using incomplete and changing information.

The proposed Flood Resilience & Response Platform addresses this problem through a connected operational workflow rather than a collection of independent tools. The platform combines a browser-based WebXR preparedness simulation, drone-based situational awareness, community reporting, geospatial state management, response coordination, route validation and shelter check-in.

The central component is the **Response Coordination Engine**. It receives the current state of affected zones and evaluates available resources and volunteers based on suitability, availability, capability, skill and distance. It then produces a recommendation for a human authority to review.

The platform deliberately uses a human-in-the-loop model. Computer vision is used to detect people and vehicles from simulated or pre-recorded drone footage. It is not presented as a reliable property-damage detector. Road and infrastructure damage can instead be reported by people in the affected area.

## 1.2 System Vision, Scope & Primary Objectives

The system vision is to convert fragmented flood information into a shared operational picture and then convert that picture into an actionable, reviewable response recommendation.

| Objective | Engineering interpretation |
|---|---|
| O1 — Preparedness | Provide a browser-based WebXR scenario that teaches users how rising water levels can affect evacuation and rescue decisions. |
| O2 — Situation Awareness | Combine drone observations and community reports into a common geospatial operational state. |
| O3 — Resource Coordination | Prioritize affected zones and recommend appropriate resources based on capability, availability and distance. |
| O4 — Volunteer Coordination | Match available citizens/volunteers to suitable response tasks based on their profile and skills. |
| O5 — Route Safety | Generate candidate routes and check them against known flood or road hazards. |
| O6 — Rescue Tracking | Record shelter check-ins so the system can represent a rescued person's safe status. |

## 1.3 Key Differentiators & Industry Alignment

The differentiation of the proposed platform is not the use of WebXR, drones, maps or machine learning individually. The differentiation comes from connecting those capabilities into a response workflow.

**1. Preparedness-to-response continuity.** The same platform covers both learning before a disaster and coordination during a disaster.

**2. Human-plus-community intelligence.** Drone observations provide one source of information while citizens can directly report conditions that computer vision may not reliably infer.

**3. Coordination as the central intelligence layer.** The system does not stop at displaying information. It uses the information to recommend who and what should respond.

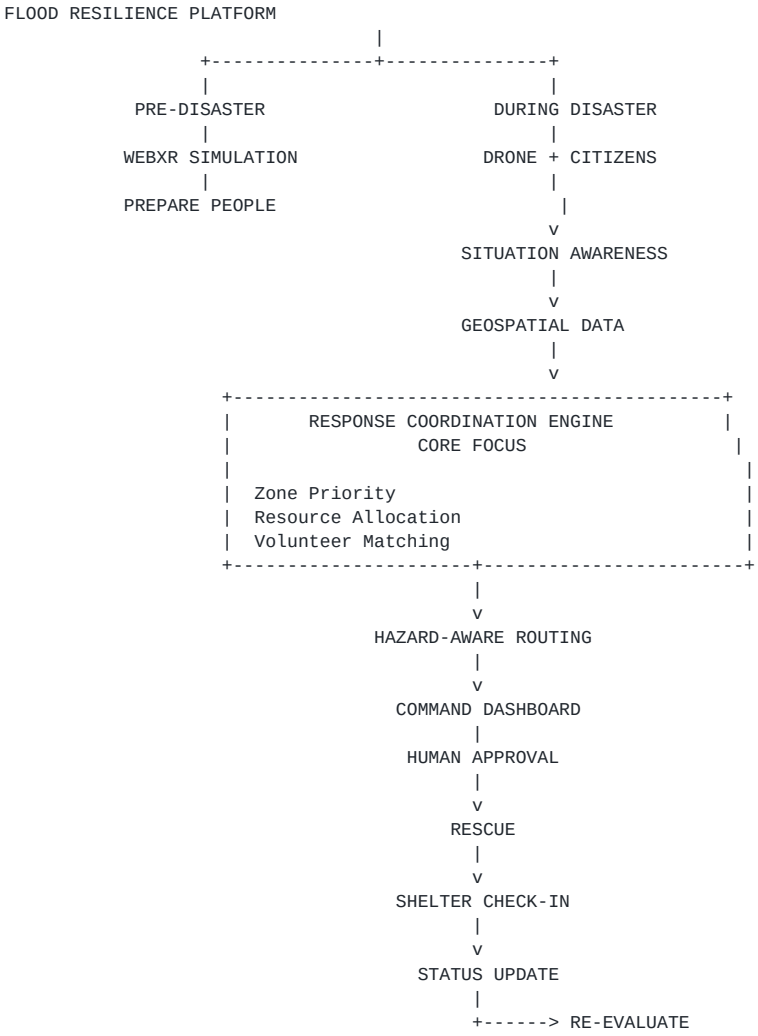
**4. Human approval.** Automated scoring does not directly dispatch rescue teams. Recommendations remain reviewable by an authorized operator.

# 2. System Architecture & Technical Design

## 2.1 Overall System Architecture & Data Flow

The architecture follows the operational sequence: **Prepare** → **Sense** → **Analyze** → **Coordinate** → **Route** → **Rescue** → **Track** → **Re-evaluate.**

The preparedness phase uses WebXR to simulate flood situations. During an actual or simulated disaster, drone observations and citizen reports contribute information to the shared geospatial state. The coordination engine uses this state to produce response recommendations.




**Core architectural principle:** computer vision provides observations; the coordination engine generates recommendations; the command dashboard provides human oversight before a response action is approved.

## 2.2 Comprehensive Technology Stack

| Layer | Technology | Purpose |
|---|---|---|
| Frontend | React | Web application and dashboards. |
| Maps | Mapbox GL JS | Map, zones, reports, shelters and routes. |
| Simulation | A-Frame / Three.js + WebXR | Browser-based flood preparedness simulation. |
| Offline Web | PWA + IndexedDB | Installable application and pending-report storage. |

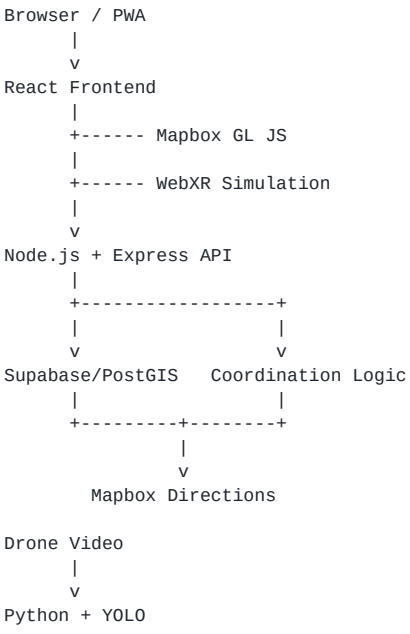
| Layer | Technology | Purpose |
|---|---|---|
| Backend | Node.js + Express | API layer and business logic. |
| AI / CV | Python + Ultralytics YOLOv8 | People and vehicle detection from drone footage. |
| Database | Supabase + PostgreSQL + PostGIS | Persistent and geospatial application data. |
| Routing | Mapbox Directions API | Candidate route generation. |
| Spatial Analysis | Turf.js | Route and hazard geometry operations. |
| Deployment | Docker | Reproducible packaging and deployment. |

## 2.3 Modular System Components

| Module | Input / Trigger | Processing | Output |
|---|---|---|---|
| WebXR Simulation | Scenario selection | Water-level and scenario progression | Training and decision feedback |
| Drone Intelligence | Drone footage | Frame processing + YOLO inference | People/vehicle observations |
| Community Reporting | Citizen report | Validation + geospatial storage | Hazard/incident report |
| Geospatial Data | Validated observations | Persistence + spatial queries | Current operational state |
| Coordination Engine | Zones + resources + profiles | Priority + suitability scoring | Recommended assignment |
| Routing | Assignment + hazards | Route generation + hazard check | Candidate safer route |
| Command Dashboard | Current state + recommendation | Human review | Approve / modify / reject |
| Shelter Tracking | Rescue/check-in | Status update | Safe / checked-in status |

## 2.4 Infrastructure & Scalability Strategy

The MVP is intentionally modular rather than unnecessarily distributed. The frontend, backend, AI processing and database have clear responsibilities while avoiding infrastructure that does not directly contribute to the demonstrated workflow.



```
      |
      v
Normalized Observations
      |
      v
Backend / PostGIS
```

Potential future scaling can include dedicated AI workers, job queues, database connection pooling, caching and more advanced offline capabilities. These are not required to demonstrate the MVP.

# 3. Repository & Directory Structure

## 3.1 Folder Hierarchy & File Layout

```
flood-resilience/
|
+-- frontend/
|   +-- src/
|   |   +-- components/
|   |   +-- pages/
|   |   +-- maps/
|   |   +-- simulation/
|   |   +-- reports/
|   |   +-- dashboard/
|   |   +-- services/
|   |   +-- types/
|   |   `-- main.tsx
|   +-- public/
|   `-- package.json
|
+-- backend/
|   +-- src/
|   |   +-- routes/
|   |   +-- controllers/
|   |   +-- services/
|   |   +-- coordination/
|   |   +-- routing/
|   |   +-- validation/
|   |   +-- database/
|   |   `-- server.ts
|   `-- package.json
|
+-- ai/
|   +-- detection/
|   |   +-- inference.py
|   |   `-- preprocessing.py
|   `-- requirements.txt
|
+-- database/
|   +-- migrations/
|   `-- seeds/
|
+-- tests/
|   +-- frontend/
|   +-- backend/
|   +-- coordination/
|   `-- routing/
|
+-- docker/
|   `-- Dockerfile
|
+-- docs/
|
`-- README.md
```

## 3.2 Key File Responsibilities

| Directory / File | Responsibility |
|---|---|
| frontend/src/simulation | WebXR scenario loading, water-level state and simulation interaction. |
| frontend/src/maps | Map rendering, markers, hazard polygons and route display. |
| frontend/src/reports | Citizen reporting forms and submission state. |
| frontend/src/dashboard | Authority view of zones, resources and recommendations. |
| backend/src/coordination | Zone priority, resource allocation and volunteer matching. |
| backend/src/routing | Route requests and hazard-intersection workflow. |
| backend/src/validation | Input validation, permissions and report normalization. |

| Directory / File | Responsibility |
| --- | --- |
| ai/detection | Drone-frame preprocessing and YOLO inference. |
| database/migrations | Version-controlled schema changes. |
| tests | Automated verification and edge-case testing. |

## 3.3 Codebase Navigation Guide

A developer debugging a citizen report should follow the path report UI → API route → validation → database persistence → geospatial state → dashboard.

A developer debugging resource allocation should follow zone state → coordination service → scoring functions → selected resource/volunteer → route service → dashboard approval.

A developer debugging drone detection should follow uploaded video → frame extraction → YOLO inference → normalized detection → API ingestion → geospatial record.

# 4. Detailed Feature Breakdown & User Workflows

## 4.1 Security & Credential Management

The platform separates citizen-facing actions from authority-level operational actions. Authentication and authorization should be handled through the selected Supabase authentication model or an equivalent backend-managed mechanism.

| Actor | Typical Permissions |
|---|---|
| Citizen | Create reports/SOS and view permitted personal or public status. |
| Volunteer | Maintain profile, availability and receive permitted assignments. |
| Authority | View operational dashboard, manage resources and approve recommendations. |
| Shelter Operator | Record authorized shelter check-ins and occupancy. |

## 4.2 Core Business Logic & Execution Engines

### 4.2.1 WebXR Flood Simulation

**Input / Trigger:** a learner selects a flood scenario and starts the simulation.

**Processing:** the scenario progresses through predefined water-level and situational states. The learner encounters decisions involving movement, evacuation and rescue response.

**Output / Action:** the interface provides feedback based on the selected action and helps the learner understand how different flood conditions affect decisions.

### 4.2.2 Drone-Based Situational Awareness

**Input / Trigger:** simulated or pre-recorded drone footage is provided to the prototype.

**Processing:** frames are extracted and YOLO is used to identify target classes such as people and vehicles.

**Output / Action:** detections become observations that can be represented on the operational map.

**Scope limitation:** property damage is not treated as a reliably solved YOLO task in the MVP. Road and infrastructure conditions can instead be supplied through community reports.

### 4.2.3 Community Reporting

**Input / Trigger:** a citizen reports an incident such as a blocked road, damaged infrastructure, rising water, stranded person or emergency request.

**Processing:** the report is validated, timestamped, geolocated and stored with its category, severity and optional evidence.

**Output / Action:** the report becomes part of the shared operational state and can influence zone priority or route safety.

### 4.2.4 Response Coordination Engine — CORE FOCUS

**Input / Trigger:** the system receives an updated operational state for one or more affected zones.

**Processing:** the engine calculates zone priority and evaluates available resources and volunteers according to capability, skill, availability, suitability and distance.

**Output / Action:** the engine produces a recommended resource and volunteer assignment for human review.

**Key principle:** the engine recommends a response. It does not independently authorize or perform the rescue.

### 4.2.5 Hazard-Aware Route Optimization

**Input / Trigger:** a resource needs to move from its current location toward an affected zone or shelter.

**Processing:** a candidate route is generated and checked against known hazard geometry such as blocked roads or severe flood zones.

**Output / Action:** a candidate safer route is shown to the authority. If unsuitable, an alternative route can be requested.

### 4.2.6 Shelter Check-In & Family Status

**Input / Trigger:** a rescued person reaches a registered shelter and is checked in.

**Processing:** the shelter record and person's status are updated while preserving access control.

**Output / Action:** authorized family members can see that the person has reached a safe shelter rather than relying only on informal communication.

# 4. Detailed Feature Breakdown & User Workflows — Continued

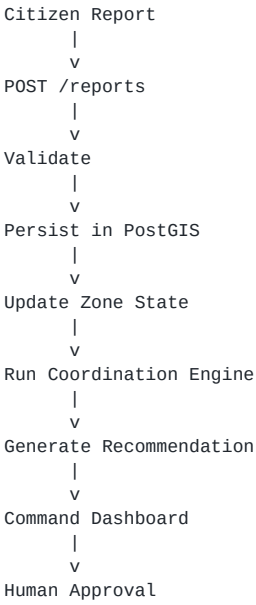
## 4.3 User Interface & Command Dashboard

The command dashboard is designed as an operational interface rather than a decorative visualization. Its purpose is to allow an authority to understand what is happening, why a recommendation was produced and what action can be approved.

| Dashboard Area | Information Displayed |
|---|---|
| Zone Status | Priority, severity, affected count, active SOS and major hazards. |
| Resource Panel | Resource type, capability, availability and location. |
| Volunteer Panel | Relevant skills, availability and suitability. |
| Recommendation | Suggested assignment and reasoning. |
| Route | Candidate route and detected hazard conflicts. |
| Approval Controls | Approve, modify or reject recommendation. |
| Shelter Panel | Capacity, occupancy and check-in status. |
| Reports Layer | Citizen reports and verification state. |

## 4.4 Real-Time Systems & Integration

The MVP does not claim a fully autonomous IoT command system. Dynamic behavior is instead achieved by treating new reports, detection results and rescue updates as state changes.




## End-to-End Operational Example

Suppose a flood affects Zone C. A drone video produces observations of several people and a submerged vehicle. A citizen reports that a nearby road is blocked.

The database now contains people/vehicle observations, a reported road hazard and the current resource inventory. The zone priority increases because the affected situation has become more urgent.

The coordination engine identifies an available suitable resource and a volunteer whose profile matches the task. A candidate route is generated and checked against the reported blocked road.

The recommendation is displayed on the command dashboard. The authority can approve or modify it. After rescue, the person is checked into a shelter and the status becomes available for authorized family lookup.

If a new road report arrives later, the route can be evaluated again rather than assuming that the previous route remains safe.

# 5. Mathematical Foundations & Quantitative Frameworks

The MVP uses transparent scoring models so that recommendations can be explained to a human operator. These equations represent the proposed framework and should be calibrated against actual implementation and test scenarios.

## 5.1 Zone Priority Model

For zone z, define a normalized priority score:

$$P_z = w_A A_z + w_C C_z + w_S S_z + w_H H_z$$

A is the affected-population measure. C represents critical cases. S represents SOS/emergency requests. H represents hazard severity. The weights determine the relative importance of each factor.

## 5.2 Resource Allocation Model

For resource r considered for zone z:

$$Q_{r,z} = w_C C_{r,z} + w_S S_{r,z} + w_A A_r - w_D D_{r,z} - w_T T_{r,z}$$

C represents capability suitability, S represents skill suitability, A represents availability, D represents distance and T represents estimated travel time.

Unavailable or incompatible resources should be filtered before scoring so that an unsuitable resource cannot be selected merely because it is nearby.

## 5.3 Volunteer Matching Model

For volunteer v and zone z:

$$V_{v,z} = w_K K_{v,z} + w_A A_v - w_D D_{v,z}$$

K represents skill/profile compatibility. A represents availability and D represents distance.

## 5.4 Route Hazard Model

For candidate route R, a hazard penalty can be represented as:

$$H(R) = \Sigma \lambda_i I_i(R)$$

I indicates whether route R intersects hazard i. λ represents the severity penalty of that hazard.

A route may be rejected or flagged if its accumulated hazard penalty exceeds a configured threshold.

## 5.5 Mathematical Summary Matrix

| Metric | Meaning | Decision Use |
|---|---|---|
| Pz | Zone priority score | Rank affected zones. |
| Qr,z | Resource suitability | Rank available resources. |
| Vv,z | Volunteer suitability | Rank suitable volunteers. |
| H(R) | Route hazard penalty | Reject or flag unsafe routes. |
| Os = Ns / Cs | Shelter occupancy ratio | Monitor shelter load. |

# 6. Database Schema & Security Infrastructure

## 6.1 Entity-Relationship Overview


The database is location-centric. Entities that influence operational decisions should carry a geographic representation or reference a geographic entity.

## 6.2 Database Tables

| Table | Important Fields | Purpose |
|---|---|---|
| users | id, role, created_at | Identity and role. |
| citizen_profiles | user_id, profile fields | Citizen information. |
| zones | id, name, severity, priority, geometry | Operational flood zones. |
| hazards | id, zone_id, type, severity, geometry | Road/flood/infrastructure hazards. |
| reports | id, user_id, type, severity, location, status | Community observations. |
| sos_requests | id, user_id, location, severity, status | Emergency requests. |
| drone_observations | id, class, confidence, location, timestamp | Computer-vision observations. |
| resources | id, type, capability, availability, location | Rescue resources. |
| volunteer_profiles | user_id, skills, availability, location | Volunteer capabilities. |
| resource_assignments | id, zone_id, resource_id, volunteer_id, status | Recommended/approved assignments. |
| shelters | id, name, capacity, occupancy, location | Shelter information. |
| shelter_checkins | id, person_id, shelter_id, status, timestamp | Rescue and shelter status. |

## 6.3 Security & Row-Level Access

Access control should ensure that citizens cannot access authority-only operational information. Volunteers should only access assignments and profile information appropriate to their role. Shelter operators should only manage records within their permitted shelter scope.

Family-facing status lookup should expose only intentionally shareable information such as a person's safe/check-in status. Internal operational information, resource locations and authority notes should remain protected.

Privileged service credentials must never be exposed in the browser. Secrets should be stored through environment variables and deployment secret management.

# 7. Local Development & Installation Guide

## 7.1 Prerequisites

| Component | Requirement |
|---|---|
| Node.js | Maintained LTS release compatible with the selected frontend/backend packages. |
| npm | Bundled with the selected Node.js release. |
| Python | 3.10+ recommended for YOLO inference. |
| Git | Current stable release. |
| Browser | Modern browser; WebXR support depends on device/browser. |
| Mapbox | Access token for map functionality. |
| Supabase | Project URL and public client key. |

## 7.2 Setup

```
git clone <repository-url>
cd flood-resilience

npm install

python -m venv .venv

# Windows
.venv\Scripts\activate

# macOS/Linux
source .venv/bin/activate

pip install -r ai/requirements.txt
```

The exact repository commands should be updated once the actual implementation repository is finalized.

## 7.3 Environment Variables

| Variable | Required | Purpose |
|---|---|---|
| VITE_MAPBOX_TOKEN | Yes | Mapbox access for the frontend. |
| SUPABASE_URL | Yes | Supabase project endpoint. |
| SUPABASE_ANON_KEY | Yes | Public client key. |
| SUPABASE_SERVICE_ROLE_KEY | Backend only | Privileged operations. Never expose to browser. |
| API_BASE_URL | Yes | Backend API address. |
| YOLO_MODEL_PATH | AI service | Selected YOLO model path. |

## 7.4 Running the Development Suite

```
# Frontend
cd frontend
npm run dev

# Backend
cd backend
npm run dev
# AI service
```

```
source .venv/bin/activate
python ai/detection/inference.py
```

Credentials must not be committed to Git. Local environment files and deployment secret stores should be used.

# 8. Verification, Testing & Quality Assurance

## 8.1 Static Verification

```
npm run lint
npm run typecheck
npm run build
npm test
```

The exact commands depend on the final package configuration. They should be included in the repository's package scripts.

## 8.2 Feature Validation

| Area | Test | Expected Result |
|---|---|---|
| WebXR | Scenario water level changes | Scenario state changes consistently. |
| YOLO | Frame contains people | People detections are produced. |
| YOLO | Frame contains vehicle | Vehicle may be flagged. |
| Community | Blocked road report | Geospatial hazard is stored. |
| Priority | Higher SOS zone | Priority increases appropriately. |
| Allocation | Nearest resource unavailable | Unavailable resource is excluded. |
| Allocation | More capable distant resource | Suitability can favor capability. |
| Routing | Route intersects hazard | Route is flagged/rejected. |
| Shelter | Person checks in | Safe status and occupancy update. |

## 8.3 Edge Cases & Operational Guardrails

| Failure State | Guardrail |
|---|---|
| No resource available | Show no-assignment condition rather than fabricate a dispatch. |
| No safe route | Flag route unavailable and require authority intervention. |
| Low-confidence YOLO result | Treat as an observation requiring review. |
| Conflicting citizen reports | Retain source and verification state. |
| Offline report | Store pending report locally and synchronize later. |
| Shelter full | Prevent normal assignment to full shelter and surface alternatives. |
| Duplicate SOS | Detect or merge duplicate requests using request/user/time rules. |
| Stale hazard | Record timestamp and surface data age to the operator. |

**Safety principle:** the platform should fail visibly. When information is missing or contradictory, it should surface uncertainty rather than silently making an unsafe operational decision.

# 9. Multi-Platform Deployment & DevOps Pipeline

## 9.1 Web Deployment

The frontend can be built as a modern web application and deployed through a static web hosting platform. Public client configuration can be provided through environment variables.

```
npm install
npm run build
```

The backend and AI inference components should remain separate from static frontend assets.

## 9.2 Containerized Deployment

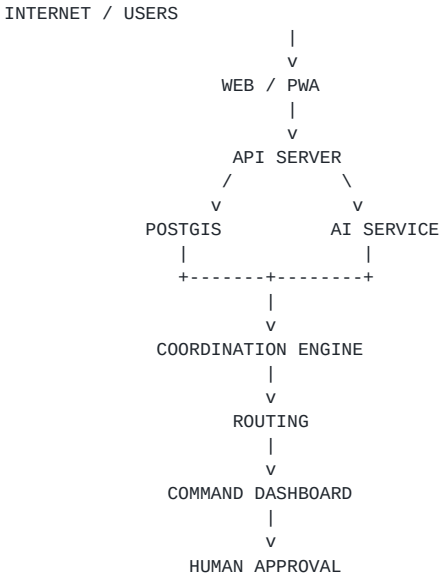
Docker can provide a reproducible runtime for the backend and AI processing components.

```
docker build -t flood-resilience-backend .
docker run --env-file .env -p 3000:3000 flood-resilience-backend
```

## 9.3 Optional Mobile Packaging

The first implementation target is web/PWA. If the project later requires a native Android package, Capacitor can be evaluated. It should not become a core dependency unless the submission requires native mobile packaging.

## Deployment Flow




The deployment architecture keeps operational decision logic separate from client presentation and AI inference.

# 10. Changelog, Roadmap, Troubleshooting & License

## 10.1 Development History

| Phase | Design Decision |
|---|---|
| Architecture Scoping | Reduced the project from disconnected technology features to a single disaster-response workflow. |
| WebXR Scoping | Moved from native AR/VR ambitions toward browser-based WebXR for feasibility. |
| Drone Scoping | Use simulated/pre-recorded drone footage for prototyping. |
| Computer Vision Scope | Use YOLO primarily for people and vehicles instead of claiming reliable property-damage detection. |
| Community Intelligence | Use citizen reports for blocked roads and infrastructure/property damage information. |
| Coordination | Defined zone-based resource allocation and volunteer matching as the core response feature. |
| Routing | Use route generation plus hazard checks rather than autonomous navigation. |
| Shelter Tracking | Add check-in status so separated families can determine whether a person reached safety. |

## 10.2 Roadmap

| Milestone | Objective |
|---|---|
| M1 | Citizen reporting and geospatial dashboard. |
| M2 | WebXR flood scenario. |
| M3 | Drone video prototype and YOLO detection. |
| M4 | Zone priority, resource allocation and volunteer matching. |
| M5 | Hazard-aware route validation and human approval. |
| M6 | Shelter check-in and family status. |
| M7 | Integrated testing and deployment. |
| Future | Optional IoT sensors, richer remote sensing, stronger offline support and native mobile packaging. |

## 10.3 Troubleshooting Matrix

| Problem | Likely Cause | Resolution |
|---|---|---|
| Map does not load | Missing/invalid Mapbox token | Check environment variable and allowed origins. |
| Report not stored | API/database configuration | Check validation, credentials and schema. |
| YOLO finds nothing | Poor footage or model mismatch | Inspect frames, target classes and confidence threshold. |
| Route crosses hazard | Hazard check missing | Run spatial intersection before approval. |
| WebXR unavailable | Browser/device limitation | Provide a normal browser simulation fallback. |
| Offline report lost | Local queue failure | Persist pending report in IndexedDB. |

## 10.4 Licensing & Intellectual Property

The final repository license should be selected by the project team. This proposal does not assert a license that has not yet been selected.

If an open-source license such as MIT is selected, the final repository should contain the complete license text and appropriate third-party dependency notices.

## 10.5 Error Log Book

| Challenge | Cause / Limitation | Design Response |
|---|---|---|
| Native AR/VR expertise unavailable | Team skill and time constraints | Use browser-based WebXR. |
| YOLO does not reliably detect property damage | Model/task mismatch | Limit CV to people/vehicles and use community reports. |
| Road damage difficult to infer | Aerial imagery may not provide reliable classification | Allow people to report blocked/damaged roads. |
| Architecture became too broad | Too many technologies introduced | Keep the MVP centered around coordination. |
| Autonomous dispatch risk | AI observation is not operational authority | Keep human approval before dispatch. |

# Appendix A — Core Technology Reference

| Technology | Project Role | MVP Status |
|---|---|---|
| React | Web application and dashboard | Core |
| Mapbox GL JS | Geospatial visualization | Core |
| A-Frame / Three.js | WebXR simulation | Core |
| WebXR | Immersive browser simulation | Core |
| PWA | Installable web experience | Core |
| IndexedDB | Offline/pending reports | Fallback |
| Node.js / Express | API and business logic | Core |
| Python / YOLOv8 | Drone computer vision | Prototype |
| Supabase | Backend platform | Core |
| PostgreSQL | Relational database | Core |
| PostGIS | Geospatial data | Core |
| Mapbox Directions | Route generation | Core |
| Turf.js | Spatial route/hazard checks | Core |
| Docker | Deployment packaging | Support |

# Appendix B — Core Coordination Pseudocode

```
function recommendResponse(zone, resources, volunteers):

    priority = scoreZone(zone)

    eligibleResources = filter(
        resources,
        resource =>
            resource.available
            AND capabilityMatches(resource, zone)
    )

    eligibleVolunteers = filter(
        volunteers,
        volunteer =>
            volunteer.available
            AND skillMatches(volunteer, zone)
    )

    rankedResources = sortByScore(
        eligibleResources,
        suitabilityScore(resource, zone)
    )

    rankedVolunteers = sortByScore(
        eligibleVolunteers,
        volunteerScore(volunteer, zone)
    )

    recommendation = {
        zonePriority: priority,
        resource: first(rankedResources),
        volunteer: first(rankedVolunteers)
    }

    return recommendation
```

The recommendation is then passed to the command dashboard. The authority can approve, modify or reject it before a rescue action is treated as authorized.

## Appendix C — Submission Integrity

This document is a technical proposal and MVP specification. Technology choices, mathematical models and architecture described as proposed should be synchronized with the actual implementation as development progresses.

The document intentionally avoids claiming that every component is already production-ready. In particular, YOLO is not claimed to solve property damage detection, routing is not autonomous navigation, and the coordination engine is not an autonomous rescue authority.

The strongest technical claim is therefore the integration workflow: preparedness through WebXR, situation awareness through drone and community inputs, coordination through zone/resource/volunteer scoring, route validation, human approval and shelter-based status tracking.

**End of Technical Submission — Version 0.1**